

Zhiyuan Li

Postdoctoral Researcher

Aalto Robot Learning

Aalto University

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Research Interests

✉ zhiyuan.li@aalto.fi

🌐 lizhyun.github.io

🔍 Google Scholar

Multi-Agent Systems, Reinforcement Learning, Robotics, Foundation Models

Education and Work Experience

Aalto Robot Learning, Aalto University

Postdoctoral Researcher

Aug 2024 – Present

Espoo, Finland

University of Electronic Science and Technology of China

Ph.D. in Computer Science and Technology

Sep 2019 – July 2024

Chengdu China

North China Electric Power University

Bachelor in Computer Science and Technology

Sep 2015 – July 2019

Beijing China

Publication

Rethinking Temporal Consistency in Video Object-Centric Learning: From Prediction to Correspondence

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Pekka Marttinen, and Joni Pajarinen.

[ICML'26] *The Forty-Third International Conference on Machine Learning, 2026.*

Learning Progress Driven Multi-Agent Curriculum

Wenshuai Zhao, Zhiyuan Li, and Joni Pajarinen.

[ICML'25] *The Forty-Second International Conference on Machine Learning, 2025.*

AgentMixer: Multi-Agent Correlated Policy Factorization

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.

[AAAI'25] *Thirty-Ninth AAAI Conference on Artificial Intelligence, 2025.*

Optimistic Multi-Agent Policy Gradient for Cooperative Tasks

Wenshuai Zhao, Yi Zhao, Zhiyuan Li, Juho Kannala, and Joni Pajarinen.

[ICML'24] *The Forty-first International Conference on Machine Learning, 2024.*

Coordination as Inference in Multi-Agent Reinforcement Learning

Zhiyuan Li, Lijun Wu, Kaile Su, Wei Wu, Yulin Jing, Tong Wu, Weiwei Duan, Xiaofeng Yue, Xiyi Tong, and Yizhou Han.

[Neural Networks'24] *Neural Networks, 2024.*

Backpropagation Through Agents

Zhiyuan Li, Wenshuai Zhao, Lijun Wu, and Joni Pajarinen.

[AAAI'24] *Thirty-Eighth AAAI Conference on Artificial Intelligence, 2024.*

Online Coordinated NFV Resource Allocation via Novel Machine Learning Techniques

Zhiyuan Li, Lijun Wu, Xiangyun Zeng, Xiaofeng Yue, Yulin Jing, Wei Wu, and Kaile Su.

[TNSM'22] *IEEE Transactions on Network and Service Management, 2022.*

Multi-agent neighborhood coordinated and holistic optimized actor-critic framework for adaptive traffic signal control

Qi Deng, Lijun Wu, Zhiyuan Li, Kaile Su, Wei Wu, and Weiwei Duan.

[Applied Intelligence] The International Journal of Research on Intelligent Systems for Real Life Complex Problems, 2025

Closed-Loop Vision-Language Planning for Multi-Agent Coordination

Zhiyuan Li, Wenshuai Zhao, and Joni Pajarinen.

[SE@AAMAS'26] The 25th International Conference on Autonomous Agents and Multiagent Systems, 2026.

Bridging the Embodiment Gap: Disentangled Cross-Embodiment Video Editing

Zhiyuan Li, Wenyan Yang, Wenshuai Zhao, Yue Ma, Yuanpeng Tu, Pekka Marttinen, and Joni Pajarinen.

[NeurIPS'26] The Fortieth Annual Conference on Neural Information Processing Systems, 2026. (Under Review)

Why Cross-Skeleton Retargeting Is Non-Identifiable: Structural Limits of Generative Motion Models

Zhiyuan Li, Wenyan Yang, Pekka Marttinen, and Joni Pajarinen.

[NeurIPS'26] The Fortieth Annual Conference on Neural Information Processing Systems, 2026. (Under Review)

Do We Really Need the Default Recipe for Video Object-Centric Learning?

Zhiyuan Li, Rongzhen Zhao, Wenyan Yang, Wenshuai Zhao, Arno Solin, Juho Kannala, Pekka Marttinen, and Joni Pajarinen.

[TPAMI] IEEE Transactions on Pattern Analysis and Machine Intelligence. (Under Review)

Multiple robots path planning based on improved artificial potential field and multi-agent reinforcement learning

Qingfeng Yao, Qifeng Zhang, Qiang Li, Zhiyuan Li, Linghan Meng, Yingzhe Sun, and Cong Wang.

[TCDS] IEEE Transactions on Cognitive and Developmental Systems. (Under Review)

Teaching Experience

Main teaching assistant for ELEC-E8125 Reinforcement Learning at the Department of Electrical Engineering and Automation, Aalto University, Finland.

Research Funding

Scholarship under China Scholarship Council, Ministry of Education of the P. R. China, EUR 16,200, 2022.

Participated in grant application preparation for the 2026 International Collaboration in High-Performance Computing call of the Research Council of Finland, with Principal Investigators Joni Pajarinen (Aalto University) and Ville Hautamaki (University of Eastern Finland).

Awards & Honors

Academic Scholarship

University of Electronic Science and Technology of China

2019-2022

Honorable Mention of Mathematical Contest in Modeling

2017

Community Activity

Conference reviewer for AAAI, ICML, NeurIPS, ICLR, ECCV, and AAMAS.

Supervision

Mentored and collaborated closely with Ph.D. researchers Wenshuai Zhao (graduated 2025) and Yi Zhao (expected graduation 2026).